

Process Characterization Report

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCR-003

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Approvals	3
Abbreviations	4
Executive summary	5
1 Introduction	6
1.1 Product and unit operation	6
1.2 Objectives	6
1.3 Regulatory and scientific basis	6
2 Prior knowledge and quality risk basis	7
2.1 Platform and prior-product knowledge	7
2.2 Quality attributes affected by the production bioreactor	7
2.3 Risk-based prioritization and parameter selection	8
3 Materials and methods	9
3.1 Scale-down model and its qualification	9
3.2 Cell line, media and culture operation	9
3.3 Analytical methods	9
3.4 Sampling plan	10
3.5 Statistical methods	10
4 Study design	11
4.1 Factors, ranges and the knowledge space	11
4.2 Screening design	12
4.3 Response-surface design	12
4.4 Univariate assessment	12
5 Results	13
5.1 Center-point performance and reproducibility	13
5.2 Screening: factor effects	13
5.3 Response-surface models	15
5.4 Mechanistic interpretation	18
6 Design space	19
7 Process capability and robustness	20
8 Parameter classification	21
9 Contribution to the control strategy	22

10 Discussion	23
11 Conclusions	24
Deviations from the plan	25
References	26
Appendix A — Screening design matrix and responses	27
Appendix B — Response-surface design matrix and responses	29
Appendix C — Full effect and coefficient tables	31
11.1 Screening effects — all responses	31
11.2 Response-surface coefficients — remaining CQAs	33
Appendix D — Analytical methods summary	35

Field	Value
Document ID	PCR-003
Document class	Process Characterization Report
Title	Process Characterization Report: Production Bioreactor (Step 3)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 2: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—

Role	Function	Name (synthetic)	Signature	Date
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ADCC, antibody-dependent cell-mediated cytotoxicity · ANOVA, analysis of variance · BLA, Biologics License Application · CCD, central-composite design · CDC, complement-dependent cytotoxicity · CPP, critical process parameter · CQA, critical quality attribute · DoE, design of experiments · DO, dissolved oxygen · DS, drug substance · GPP, general process parameter · HCP, host-cell protein · ICH, International Council for Harmonisation · IPC, in-process control · KPP, key process parameter · LRV, log-reduction value · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · QTPP, quality target product profile · RSM, response-surface methodology · SDM, scale-down model · SEC, size-exclusion chromatography · VCD, viable cell density · WC-CPP, well-controlled CPP.

Executive summary

This report documents the Stage 1 (Process Design) characterization of the A-Mab production bioreactor (Step 3), the fed-batch mammalian cell-culture operation that establishes the N-glycosylation, charge-variant and initial size-variant and process-impurity profile of the drug substance. The study was executed under the Process Characterization Plan **PCP-003** on a qualified 2 L scale-down model (SDM), following the lifecycle approach to process validation (U.S. Food and Drug Administration 2011; Parenteral Drug Association 2013; International Society for Pharmaceutical Engineering 2023) and the Quality-by-Design principles of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012), with the process model, quality-attribute framework and risk methodology of the A-Mab case study (CMC Biotech Working Group 2009).

Nine process parameters were evaluated. A 19-run resolution-V fractional-factorial screening design identified the significant factors and two-factor interactions for five cell-culture CQAs; a 28-run face-centred central-composite response-surface design then quantified main effects, interactions and curvature and defined the multivariate design space. The response-surface models are adequate for all five CQAs ($R^2 = 0.91\text{--}0.97$, adjusted $R^2 = 0.81$, predicted $R^2 = 0.60$) with no significant lack of fit relative to the center-point pure error.

Key outcomes

- Culture pH, temperature, dissolved CO₂, osmolality and culture duration significantly affect the glycan and charge-variant CQAs and are classified **well-controlled CPP (WC-CPP)**; dissolved oxygen, initial viable cell concentration and nutrient feed-1 volume are **KPP**; basal medium concentration is **GPP**.
- The nominal fed-batch reaches a peak viable cell density of **13.48 × 10⁶ cells/mL** and a harvest titer of **4.37 g/L** at **67.1%** final viability.
- A multivariate design space in culture pH, temperature, culture duration and dissolved CO₂ was established over which all cell-culture CQAs are satisfied simultaneously.
- Across a Monte-Carlo simulation of 2,000 commercial-scale batches, all bioreactor-set CQAs meet their acceptance criteria with a minimum capability of **Cpk = 1.94**.

The production bioreactor is demonstrated to be well understood and robust. The design space, parameter classifications and proven acceptable ranges support the Stage 2 operating ranges and the drug-substance control strategy, and this report rolls up into the Process Characterization Master Report (**PCMR-001**).

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody-dependent cell-mediated cytotoxicity (ADCC), with a contributory complement-dependent cytotoxicity (CDC) component; both are governed by the Fc N-glycosylation profile (CMC Biotech Working Group 2009). The production bioreactor is a fed-batch mammalian cell culture operated at 15,000 L working volume. It is the only upstream operation that forms product-quality CQAs and is therefore the sole upstream step used to define the upstream design space; the seed-expansion and harvest operations influence titer and process consistency but do not form or modify product-quality attributes (CMC Biotech Working Group 2009). Product quality established in the bioreactor — afucosylation, high mannose, galactosylation, acidic charge variants and aggregate, together with the host-cell protein (HCP) and DNA impurity load presented to the purification train — is the subject of this characterization.

1.2 Objectives

The objectives of the study were to (i) establish the quantitative relationships between the production-bioreactor process parameters and the cell-culture CQAs; (ii) define the multivariate design space over which all CQAs are met simultaneously; (iii) establish proven acceptable ranges (PARs) and normal operating ranges (NORs) for each parameter; (iv) classify each parameter for its risk of impact to CQAs and to process performance; and (v) provide the process understanding and ranges that justify the Stage 2 operating conditions and the control strategy.

1.3 Regulatory and scientific basis

The work follows the lifecycle approach to process validation of the FDA 2011 guidance (U.S. Food and Drug Administration 2011), PDA Technical Report 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023), and applies the enhanced (QbD) development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012). Criticality is treated as a continuum (ICH Q9). The report structure corresponds to the Stage 1 Process Design Report of PDA TR 60 §3.11 and provides development-report content suitable to support a Biologics License Application.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The characterization ranges and set-points are supported by platform knowledge from related humanized IgG1 antibodies (referred to as X-Mab, Y-Mab and Z-Mab) and by A-Mab clinical and engineering manufacturing experience (CMC Biotech Working Group 2009). Prior knowledge established that Fc glycosylation (afucosylation, high mannose, galactosylation) governs ADCC/CDC potency and is set in cell culture; that aggregate formed in culture is subsequently polished downstream; and that the harvest HCP and DNA burden is generated upstream and cleared by the purification train. This knowledge framed the CQAs in scope and the prioritization of parameters for characterization.

2.2 Quality attributes affected by the production bioreactor

Table 2.1 summarizes the CQAs set or principally established by the production bioreactor, with their acceptance criteria and criticality. Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact $\times$ Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009).

Table 2.1: CQAs established by the production bioreactor, with acceptance criteria and criticality (Tool #1 = Impact $\times$ Uncertainty; H/VH = Critical).

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

The glycosylation attributes are the most critical because they govern potency; afucosylation and galactosylation are managed within target ranges, whereas high mannose (Very High criticality) is controlled to a narrow upper limit. Aggregate is established in culture and polished at cation

exchange; acidic charge variants, HCP and residual DNA are followed here as they load the purification train.

2.3 Risk-based prioritization and parameter selection

Consistent with the A-Mab lifecycle of iterative risk assessments, the characterization scope was set by the Pre-Characterization Process Risk Assessment (**RA-001**), which used prior knowledge and a risk-ranking-and-filtering screen to prioritize parameters and to assign the study type. Parameters with a credible main-effect and interaction risk to the glycan and charge-variant CQAs (culture pH, temperature, dissolved CO₂, osmolality and culture duration) were assigned to the multivariate DoE; parameters expected to affect only process performance were assigned to univariate assessment or held at their set-point (CMC Biotech Working Group 2009; Parenteral Drug Association 2013). The parameters and ranges carried into this study are given in Table 4.1 (§5.1).

3 Materials and methods

3.1 Scale-down model and its qualification

All characterization runs were performed on a qualified 2 L bench-scale bioreactor operated at the commercial working-volume equivalent for pH, temperature, dissolved oxygen and CO control, gas sparging, agitation power per unit volume and feeding strategy (**SOP-2003**, **SOP-2004**). The SDM was qualified against at-scale reference data (**SOP-1001**) by comparing both the operational parameters and, critically, the input and output quality attributes — viable cell density, viability, titer, and the afucosylation, galactosylation, high-mannose, charge-variant, aggregate, HCP and DNA profiles — confirming that the model is representative and suitable to support design-space and PAR claims (Parenteral Drug Association 2013). The between-run reproducibility of the center-point condition (§6.1) provides a direct, quantitative estimate of model precision and is used as the pure-error term in the lack-of-fit assessment of every response-surface model.

3.2 Cell line, media and culture operation

The production culture is a fed-batch process inoculated from the N-1 seed bioreactor at the target initial viable cell concentration, using platform basal medium and a bolus nutrient feed (**SOP-2004**). Culture pH, temperature, dissolved oxygen and dissolved CO are controlled to set-point by the bioreactor control system; osmolality is a monitored culture attribute governed by the feed and base addition; and the culture is harvested on the nominal duration criterion. Raw materials and cell-culture media are controlled to platform specifications and are treated as identity/quality-controlled inputs rather than characterized variables, consistent with the A-Mab platform approach (CMC Biotech Working Group 2009).

3.3 Analytical methods

CQAs were measured by validated methods (Table 3.1). Released N-glycans were determined by 2-AB HILIC-UPLC (**AMV-3010**); size variants (aggregate/HMW) by SEC-HPLC (**AMV-3011**); HCP by ELISA (**AMV-3012**); charge variants by imaged capillary isoelectric focusing (**AMV-3013**); and residual host-cell DNA by qPCR (**AMV-3014**). Method performance characteristics are summarized in Appendix D.

Table 3.1: Validated analytical methods used in the characterization.

Reference	Title	Type
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

3.4 Sampling plan

Each DoE run was sampled at harvest for the released CQAs (glycan map, charge variants, aggregate) and for the harvest HCP and residual-DNA load; daily process samples (viable cell density, viability, metabolites, pH/pCO₂ verification) were taken to confirm culture trajectory. Center-point runs were replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

3.5 Statistical methods

Designs were constructed and analyzed in coded factors, with each parameter’s characterization range mapped so that the set-point is 0 and the range edges are ± 1 (**SOP-1002**). Screening data were analyzed by ordinary-least-squares regression of each response on the main effects and all two-factor interactions; factor *effects* are reported as twice the coded coefficient. Response-surface data were analyzed with the full quadratic model (main effects, two-factor interactions and pure-quadratic terms). Statistical significance was assessed at $\alpha = 0.05$. Model adequacy was judged by the coefficient of determination (R^2), the adjusted R^2 , the predicted R^2 (computed from the PRESS statistic by leave-one-out), the overall model F-test, and an analysis of variance partitioning the residual into lack of fit and pure error, the latter estimated from the replicated center points; a non-significant lack-of-fit F-test ($p > 0.05$) indicates an adequate model. The design space was defined as the multivariate region of the four dominant factors over which every CQA response satisfies its acceptance criterion simultaneously. Commercial-scale process capability was estimated by Monte-Carlo simulation of 2,000 batches with each parameter varied within its NOR, reporting the process-capability index (Cpk) one-sided for impurity CQAs and two-sided for target-range CQAs.

4 Study design

4.1 Factors, ranges and the knowledge space

Nine parameters were studied (Table 4.1). The characterization range of each multivariate factor (“PAR” column) defines the edges of the explored knowledge space and the basis of the proven acceptable range; the NOR is the tighter range at which the parameter is controlled routinely. The DoE factor coding is given in Table 4.2.

Table 4.1: Production-bioreactor process parameters, set-points, ranges and post-characterization classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP	multivariate
Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP	multivariate
Dissolved CO ₂ (pCO ₂)	mmHg	100	70–130	40–160	WC-CPP	multivariate
Osmolality	mOsm	400	375–425	360–440	WC-CPP	multivariate
Culture duration	days	17	16–18	15–19	WC-CPP	multivariate
Dissolved oxygen	% sat	50	40–60	30–70	KPP	multivariate
Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	KPP	univariate
Basal medium concentration	X	1.2	1–1.4	0.8–1.6	GPP	multivariate
Nutrient feed-1 volume	% WV	12	10–14	9–15	KPP	multivariate

Table 4.2: Design-of-experiments factor legend (coded A–E).

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO ₂ (pCO ₂)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

4.2 Screening design

Factor screening used a resolution-V two-level fractional factorial in the five multivariate factors (A–E) with three center-point replicates, 19 runs in total. At resolution V, main effects are unconfounded with one another and with two-factor interactions, so both can be estimated cleanly. The five responses recorded for each run were afucosylation, galactosylation, high mannose, acidic charge variants and aggregate. The complete coded design matrix and run responses are given in Appendix A.

4.3 Response-surface design

The four factors carried forward from screening (culture pH, temperature, culture duration and dissolved CO₂ — codes A, B, E, C) were studied in a face-centred central-composite design (cube points, axial points at the face centres, and four center-point replicates), 28 runs in total, enabling estimation of curvature in addition to main effects and interactions. The complete coded design matrix and responses are given in Appendix B.

4.4 Univariate assessment

Initial viable cell concentration was evaluated one factor at a time across its characterization range for its effect on peak viable cell density, titer and culture trajectory; it was not carried into the multivariate design because prior knowledge indicated no credible CQA impact.

5 Results

5.1 Center-point performance and reproducibility

The nominal (center-point) fed-batch profile (Figure 5.1) reaches a peak viable cell density of 13.48×10^6 cells/mL and a harvest titer of 4.37 g/L at 67.1% final viability at the nominal 17-day harvest. The between-replicate reproducibility of the center point (Table 5.1) is the pure-error estimate used in the lack-of-fit assessment; the coefficients of variation (4.1–9.9%) reflect the combined analytical and biological variability of the SDM.

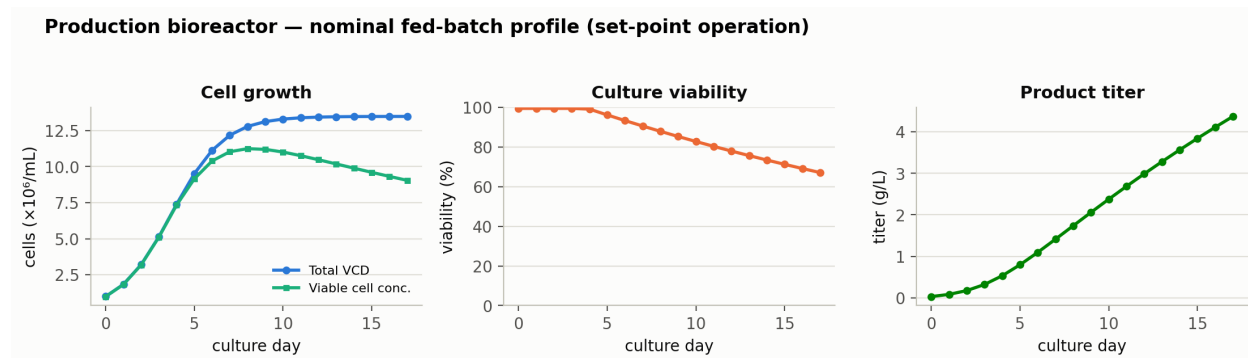


Figure 5.1: Nominal production-bioreactor time course: viable cell density, viability and titer.

Table 5.1: Center-point reproducibility (response-surface design; $n = 4$ replicates).

Response	n	Mean	SD	%CV
Afucosylation	4	7.61	0.355	4.66
Galactosylation	4	25.9	2.34	9.02
High mannose	4	5.66	0.559	9.88
Acidic variants	4	21.5	0.877	4.08
Aggregates (HMW)	4	1.51	0.0985	6.53

5.2 Screening: factor effects

The screening models explain the majority of the response variance and identify the active factors and interactions (Table 5.2). Consistent with the fractional design’s limited residual degrees of freedom, screening is used to *rank and select* effects; the magnitudes are refined by the response-surface models (§6.3).

Table 5.2: Screening-design model summary by response (main effects + two-factor interactions).

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Afucosylation	19	0.965	0.79	-47.4	5.51	0.0926	1.06
Galactosylation	19	0.998	0.988	-1.33	103	0.00138	0.773
High mannose	19	0.942	0.65	-7.99	3.23	0.182	0.619
Acidic variants	19	0.994	0.963	-0.442	32.2	0.00766	0.957
Aggregates (HMW)	19	0.995	0.972	-2.44	42.2	0.00515	0.0829

The standardized effects for the two potency-determining glycan CQAs are shown in Figure 5.2; the largest effects are tabulated for afucosylation (Table 5.3) and galactosylation (Table 5.4). Full effect tables for all five responses are given in Appendix C.

Bioreactor screening DoE — standardized effects (red = significant, $p < 0.05$)

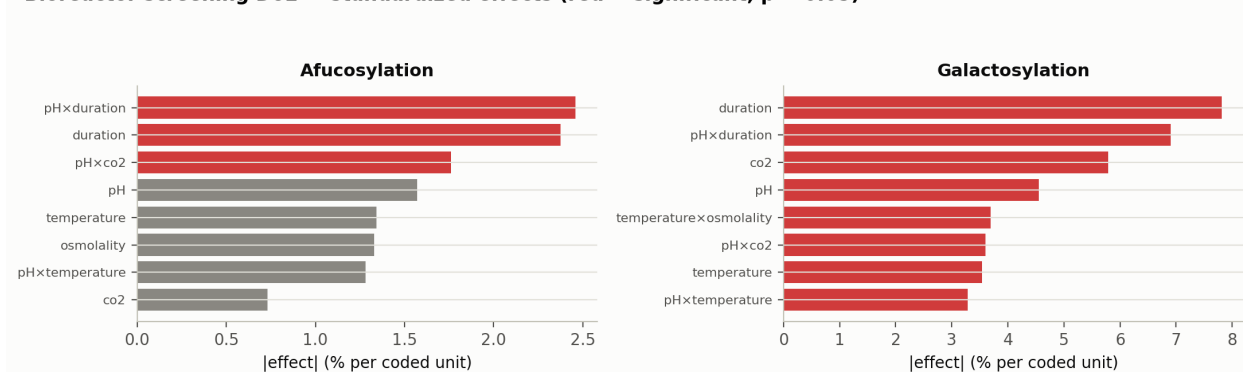


Figure 5.2: Standardized screening effects for afucosylation and galactosylation; red bars are significant ($p < 0.05$).

Table 5.3: Largest standardized effects on afucosylation (screening; effect = $2 \times$ coded coefficient).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AE	2.46	1.23	0.266	4.62	0.0191	*
E	-2.37	-1.19	0.266	-4.46	0.0209	*
AC	-1.76	-0.88	0.266	-3.31	0.0455	*
A	-1.57	-0.785	0.266	-2.95	0.06	
B	-1.34	-0.671	0.266	-2.53	0.0858	
D	-1.33	-0.664	0.266	-2.5	0.0878	
AB	-1.28	-0.64	0.266	-2.41	0.0952	
C	-0.73	-0.365	0.266	-1.37	0.264	

Table 5.4: Largest standardized effects on galactosylation (screening).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	-7.82	-3.91	0.193	-20.2	0.000264	***
AE	6.9	3.45	0.193	17.9	0.000382	***
C	-5.79	-2.9	0.193	-15	0.000644	***
A	-4.56	-2.28	0.193	-11.8	0.00131	**
BD	3.7	1.85	0.193	9.57	0.00242	**
AC	3.6	1.8	0.193	9.32	0.00261	**
B	3.54	1.77	0.193	9.17	0.00274	**
AB	3.28	1.64	0.193	8.49	0.00343	**

Culture pH, temperature, dissolved CO₂, osmolality and culture duration are the active factors across the glycan and charge-variant responses, with several significant two-factor interactions — notably culture pH $\times$ culture duration and culture pH $\times$ dissolved CO₂ — confirming that the CQAs cannot be controlled by one-factor-at-a-time set-points and motivating the multivariate response-surface characterization.

5.3 Response-surface models

The full quadratic response-surface models are adequate for all five CQAs (Table 5.5): $R^2 = 0.91$ – 0.97 , adjusted $R^2 = 0.81$, and predicted $R^2 = 0.60$, indicating good predictive capability rather than overfitting. For every response the lack-of-fit F-test was not significant ($p > 0.05$) relative to the center-point pure error, confirming model adequacy.

Table 5.5: Response-surface model summary by response (full quadratic model).

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Afucosylation	28	0.963	0.923	0.795	24.3	5.05e-07	0.584
Galactosylation	28	0.971	0.94	0.915	31.3	1.06e-07	1.36
High mannose	28	0.911	0.815	0.65	9.48	0.000118	0.472
Acidic variants	28	0.938	0.872	0.598	14.1	1.27e-05	1.57
Aggregates (HMW)	28	0.969	0.936	0.835	29.3	1.58e-07	0.0985

The fitted coefficients for afucosylation and galactosylation are given in Table 5.6 and Table 5.7, and the analysis of variance for afucosylation — including the lack-of-fit/pure-error partition — in Table 5.8. Coefficient tables for the remaining CQAs are in Appendix C.

Table 5.6: Response-surface coefficients for afucosylation (coded factors A = pH, B = temperature, C = CO₂, E = duration).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	8.04	0.202	39.8	5.74e-15	***
A	-0.997	0.138	-7.24	6.52e-06	***
B	-0.551	0.138	-4	0.00151	**
E	-1.3	0.138	-9.45	3.45e-07	***
C	-0.367	0.138	-2.67	0.0194	*
AB	-0.63	0.146	-4.32	0.000837	***
AE	1.32	0.146	9.06	5.6e-07	***
AC	-0.907	0.146	-6.21	3.16e-05	***
BE	0.362	0.146	2.48	0.0277	*
BC	-0.0703	0.146	-0.481	0.638	
EC	0.0322	0.146	0.221	0.829	
A ²	-1.37	0.364	-3.77	0.00233	**
B ²	-0.157	0.364	-0.431	0.673	
E ²	0.257	0.364	0.705	0.493	
C ²	0.117	0.364	0.323	0.752	

Table 5.7: Response-surface coefficients for galactosylation.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	25.8	0.47	54.8	9.11e-17	***
A	-2.21	0.321	-6.89	1.09e-05	***
B	2.3	0.321	7.18	7.13e-06	***
E	-3.82	0.321	-11.9	2.27e-08	***
C	-2.43	0.321	-7.56	4.11e-06	***
AB	1.67	0.34	4.92	0.000279	***
AE	2.72	0.34	7.99	2.28e-06	***
AC	2.35	0.34	6.9	1.09e-05	***
BE	-0.217	0.34	-0.639	0.534	
BC	0.315	0.34	0.928	0.37	
EC	0.332	0.34	0.975	0.347	
A ²	-0.837	0.847	-0.988	0.341	
B ²	0.378	0.847	0.446	0.663	
E ²	0.937	0.847	1.11	0.289	
C ²	-0.373	0.847	-0.441	0.667	

Table 5.8: Analysis of variance for the afucosylation response-surface model, with lack-of-fit and pure-error partition.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	116	14	8.27	24.3	5.05e-07

Source	Sum sq.	df	Mean sq.	F	p-value
Residual	4.43	13	0.341	nan	nan
Lack of fit	4.06	10	0.406	3.23	0.182
Pure error	0.377	3	0.126	nan	nan

The response surfaces over the two most influential factors (culture pH and culture duration) are shown for all five CQAs in Figure 5.3, and the model diagnostics for the afucosylation and aggregate models in Figure 5.4 and Figure 5.5. The residual-versus-predicted and normal-quantile plots show no systematic structure or departure from normality, and the actual-versus-predicted plots lie on the identity line, supporting the use of the models for design-space definition.

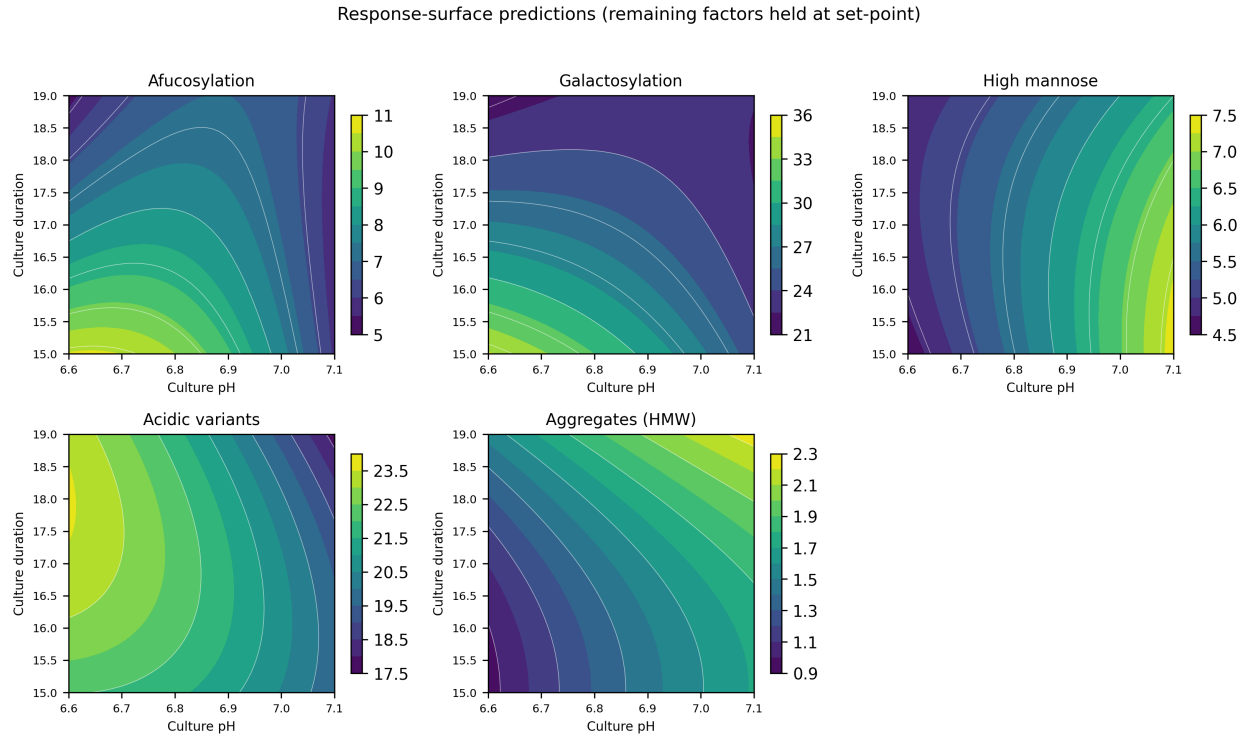


Figure 5.3: Response-surface predictions for the five cell-culture CQAs over culture pH $\times$ culture duration (remaining factors at set-point).

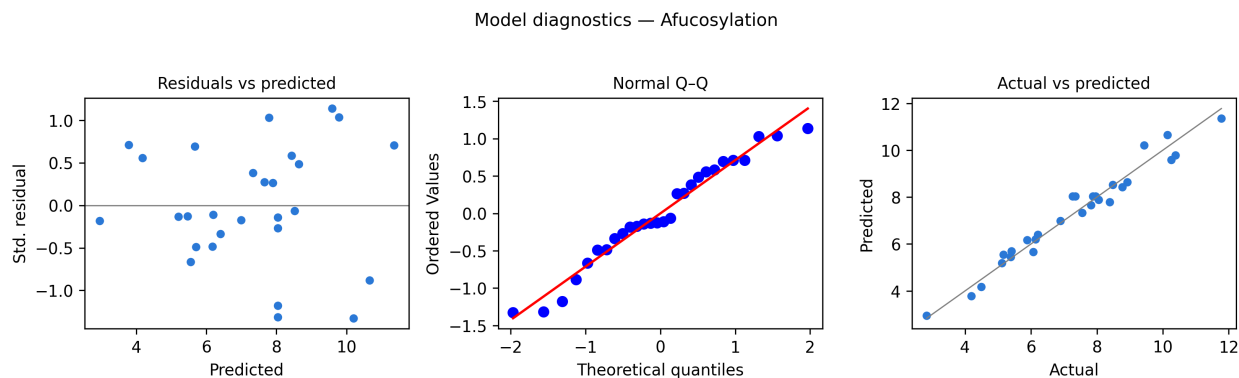


Figure 5.4: Model diagnostics for the afucosylation response-surface model.

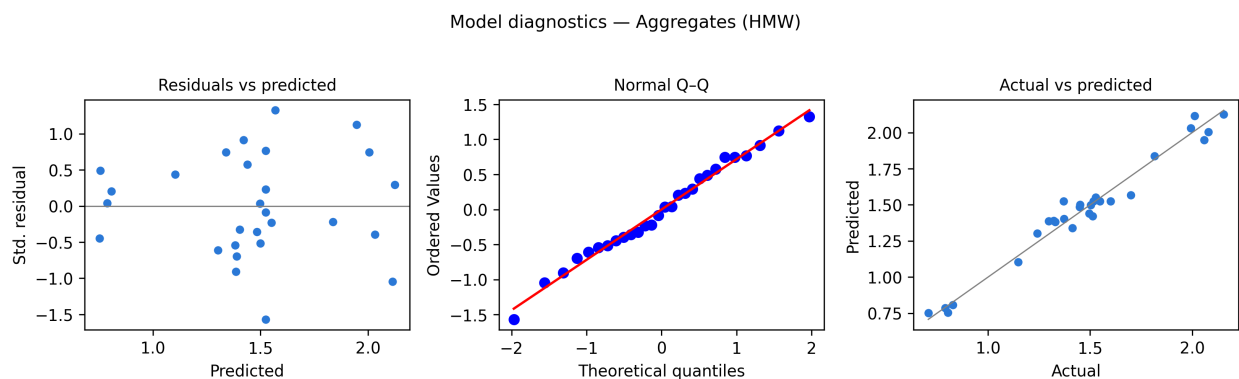


Figure 5.5: Model diagnostics for the aggregate (HMW) response-surface model.

5.4 Mechanistic interpretation

The response-surface models reproduce the established physiology of fed-batch IgG1 culture. Lower culture pH and lower temperature increase specific productivity, extend the productive culture and raise titer, with a modest increase in afucosylation and a decrease in galactosylation. Extended culture duration lowers both afucosylation and galactosylation and raises the harvest HCP and DNA load as viability declines late in culture. Dissolved CO₂ and osmolality modulate galactosylation and the charge-variant profile. The significant culture pH $\times$ duration and pH $\times$ CO₂ interactions mean the glycan responses depend on the *combination* of set-points, which is why a multivariate design space — rather than independent one-dimensional ranges — is required to assure quality.

6 Design space

The design space is the multivariate region of culture pH, temperature, culture duration and dissolved CO₂ over which all five cell-culture CQAs simultaneously meet their acceptance criteria, computed by overlaying the individual CQA constraints on the fitted response-surface models. Figure 6.1 shows the pH $\times$ culture-duration plane at the set-points of the remaining factors: the shaded region satisfies every CQA at once, with the afucosylation contours overlaid and the NOR and set-point indicated. The NOR of each parameter (Table 4.1) lies well inside the design space, which in turn lies within the broader characterized knowledge space; movement within the design space is not a change, whereas movement outside it would require assessment under the quality system (CMC Biotech Working Group 2009; International Council for Harmonisation 2009).

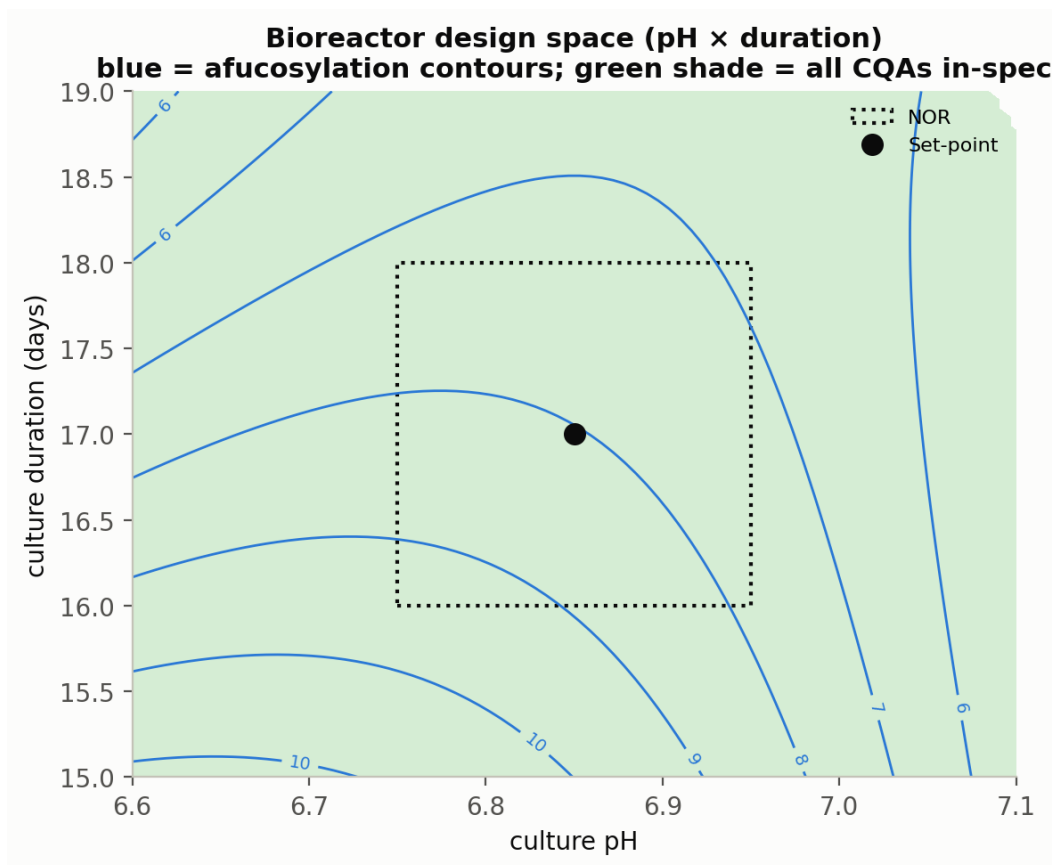


Figure 6.1: Production-bioreactor design space (culture pH $\times$ culture duration); the shaded region satisfies all cell-culture CQAs simultaneously, with afucosylation (%) contours overlaid.

7 Process capability and robustness

A Monte-Carlo simulation of 2,000 commercial-scale batches, with every parameter varied within its NOR, was used to estimate the commercial-scale CQA distributions and process capability (Table 7.1). Capability is evaluated one-sided for impurity CQAs (upper limit) and two-sided for target-range CQAs.

Table 7.1: Commercial-scale process capability for the bioreactor-set CQAs.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Afucosylation	H	two_sided	2–13	7.98 ± 0.58	2.89
Galactosylation (total %Gal)	H	two_sided	10–40	26 ± 1.6	2.86
High Mannose	VH	two_sided	3–10	5.99 ± 0.51	1.94
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.11
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14

All bioreactor-set CQAs meet their acceptance criteria with margin; the minimum capability is **Cpk = 1.94**, driven by high mannose, which is the Very-High-criticality attribute held to the narrowest target range. The capability margin confirms that the NOR-bounded process reliably delivers in- specification drug substance at commercial scale.

8 Parameter classification

Each parameter was classified on the A-Mab continuum from the demonstrated effects and the design-space risk (Table 8.1). The rationale is as follows.

- **Culture pH, temperature, dissolved CO₂, osmolality, culture duration — WC-CPP.** Each significantly affects one or more glycan and/or charge-variant CQAs (§6.2–6.3), and several participate in significant interactions. They are, however, reliably held within the design space by closed-loop control and continuous in-process monitoring, giving a low risk of leaving the design space; they are therefore well-controlled CPPs rather than CPPs.
- **Dissolved oxygen, initial viable cell concentration, nutrient feed-1 volume — KPP.** These affect peak viable cell density, titer and culture consistency but do not significantly impact the CQAs; they are controlled for process performance.
- **Basal medium concentration — GPP.** No meaningful impact on CQAs or performance was observed over the wide range studied.

Table 8.1: Post-characterization parameter classification.

Parameter	Class	Study
Culture pH	WC-CPP	multivariate
Culture temperature	WC-CPP	multivariate
Dissolved CO ₂ (pCO ₂)	WC-CPP	multivariate
Osmolality	WC-CPP	multivariate
Culture duration	WC-CPP	multivariate
Dissolved oxygen	KPP	multivariate
Initial viable cell conc.	KPP	univariate
Basal medium concentration	GPP	multivariate
Nutrient feed-1 volume	KPP	multivariate

9 Contribution to the control strategy

The production bioreactor is the primary control point for the glycosylation CQAs (afucosylation, high mannose, galactosylation) and for the acidic charge-variant profile, and it sets the initial aggregate and the HCP/DNA load presented downstream. Its contribution to the overall control strategy comprises: control of the five WC-CPPs to their NORs within the established design space; in-process monitoring of pH, temperature, dissolved oxygen and pCO₂; harvest on the qualified duration criterion; released glycan-map, charge-variant, aggregate and HCP testing; and the trending of the harvest impurity load. The glycan CQAs are controlled predominantly here because they are not modified downstream, whereas aggregate and HCP established here are further controlled by cation exchange and by the Protein A / CEX / AEX train respectively (reported in the paired unit-operation reports and consolidated in **PCMR-001**).

10 Discussion

The characterization quantifies the parameter–CQA relationships for the production bioreactor, defines a robust multivariate design space, and classifies every parameter with a data-supported rationale. The response-surface models are adequate and predictive, and their behaviour is consistent with established fed-batch IgG1 physiology and with the A-Mab prior knowledge, increasing confidence that the SDM is representative of commercial scale. The principal limitation is inherent to scale-down characterization: absolute values and the harvest impurity load may shift modestly at scale, which is why the design space is defined on the qualified SDM and confirmed at commercial scale during Stage 2, and why the WC-CPP ranges are carried forward with in-process monitoring. The characterization did not identify any parameter requiring the most stringent (CPP, as opposed to WC-CPP) control within the studied ranges.

11 Conclusions

The A-Mab production bioreactor is well understood and robust. The relationships between the culture parameters and the glycan, charge-variant and aggregate CQAs are quantified by DoE; a multivariate design space is established; every parameter range is justified by data; and all CQAs meet acceptance at commercial scale with a minimum capability of $Cpk = 1.94$. The design space, parameter classifications and proven acceptable ranges provide the basis for the Stage 2 operating ranges and the drug-substance control strategy. This report rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment.

Deviations from the plan

No deviations from **PCP-003** were recorded. All planned runs were executed and all planned responses were measured on the qualified scale-down model.

References

- CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS.
- International Council for Harmonisation. 2009. “ICH Q8(R2): Pharmaceutical Development.” ICH.
- . 2012. “ICH Q11: Development and Manufacture of Drug Substances.” ICH.
- . 2023. “ICH Q9(R1): Quality Risk Management.” ICH.
- International Society for Pharmaceutical Engineering. 2023. “Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation.” ISPE.
- Parenteral Drug Association. 2013. “Technical Report No. 60: Process Validation — a Lifecycle Approach.” PDA.
- U.S. Food and Drug Administration. 2011. “Guidance for Industry: Process Validation — General Principles and Practices.” FDA.

Appendix A — Screening design matrix and responses

The coded resolution-V fractional-factorial design (factors A–E per Table 4.2; coded levels $-1/0/+1$) and the measured responses for each run.

Table 11.1: Appendix A1 — screening design (coded factor settings).

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

Table 11.2: Appendix A2 — screening run responses.

Run	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
1	5.18	27.9	6.61	26.4	1.3
2	8.93	36.1	6.44	26.5	0.783
3	11.8	34.6	5.63	22.3	0.785
4	4.83	14.7	5.46	17.9	1.48
5	9.72	41.4	4.61	27	0.859
6	4.68	27.3	4.68	27.6	1.26

Run	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
7	6.22	14.5	4.65	23.1	1.48
8	9.83	31.2	4.45	19.3	0.948
9	9.61	28	7.77	24.1	1.51
10	7.51	17	7.89	20.4	2.26
11	7.13	19.5	7.34	12.6	2.53
12	5.29	17.6	7.01	11.3	1.61
13	6.74	28.3	6.43	29.6	2.18
14	5.4	26.7	5.94	24.4	1.62
15	3.82	25.2	6.1	18.4	1.8
16	3.08	29.1	6.15	15.8	2.05
17	7.8	26.4	5.88	21.9	1.64
18	7.51	27.4	5.84	20	1.63
19	8.48	27	7.11	21.8	1.5

Appendix B — Response-surface design matrix and responses

The coded face-centred central-composite design (factors A, B, C, E; coded levels $-1/0/+1$) and the measured responses for each run.

Table 11.3: Appendix B1 — response-surface design (coded factor settings).

Run	Type	A	B	C	E
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	1	-1
3	factorial	-1	-1	-1	1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	1	-1
7	factorial	-1	1	-1	1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	1	-1
11	factorial	1	-1	-1	1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	1	-1
15	factorial	1	1	-1	1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	0	-1
22	axial	0	0	0	1
23	axial	0	0	-1	0
24	axial	0	0	1	0
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 11.4: Appendix B2 — response-surface run responses.

Run	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
1	9.43	38.8	5.44	29.8	0.827
2	11.8	29	5.55	17.6	0.791
3	4.5	25.9	5.65	25.1	1.33
4	5.38	16.6	5.38	20	1.37
5	10.4	40.5	4.23	25.6	0.803
6	10.1	30.7	4.17	19.8	0.707
7	5.12	27.1	4.63	30.3	1.24
8	6.13	17.4	4.64	21.8	1.41
9	8.92	22	6.79	22.9	1.5
10	5.89	18.7	8.81	13.6	1.53
11	8.05	18.7	6.11	19.3	1.99
12	5.17	18.7	7.34	12.3	2.15
13	5.41	28.7	6.97	28.5	1.45
14	2.84	29.1	7.36	18.5	1.7
15	6.2	25.4	5.95	26.5	2.08
16	4.19	27.2	6.8	17.2	2.01
17	7.82	26.1	4.59	24.2	1.15
18	6.07	23.7	7.72	19.3	1.82
19	8.77	24.3	6.75	19.9	1.51
20	7.55	27.9	5.51	26.8	1.32
21	10.3	30.8	6.11	20.8	1.3
22	6.89	22.5	5.54	22.4	2.06
23	8.48	27.1	6.39	25.7	1.5
24	8.39	23.5	5.94	19.2	1.45
25	7.88	23	5.75	20.5	1.52
26	7.95	27.3	6.32	22	1.6
27	7.35	25	5.6	22.4	1.55
28	7.27	28.2	4.96	21.1	1.37

Appendix C — Full effect and coefficient tables

11.1 Screening effects — all responses

Afucosylation

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AE	2.46	1.23	0.266	4.62	0.0191	*
E	-2.37	-1.19	0.266	-4.46	0.0209	*
AC	-1.76	-0.88	0.266	-3.31	0.0455	*
A	-1.57	-0.785	0.266	-2.95	0.06	
B	-1.34	-0.671	0.266	-2.53	0.0858	
D	-1.33	-0.664	0.266	-2.5	0.0878	
AB	-1.28	-0.64	0.266	-2.41	0.0952	
C	-0.73	-0.365	0.266	-1.37	0.264	
BD	0.452	0.226	0.266	0.85	0.458	
BE	0.362	0.181	0.266	0.681	0.545	
AD	-0.175	-0.0876	0.266	-0.329	0.764	
BC	-0.171	-0.0856	0.266	-0.322	0.769	
CD	-0.147	-0.0736	0.266	-0.277	0.8	
DE	0.0358	0.0179	0.266	0.0674	0.951	
CE	0.0118	0.00591	0.266	0.0222	0.984	

Galactosylation

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	-7.82	-3.91	0.193	-20.2	0.000264	***
AE	6.9	3.45	0.193	17.9	0.000382	***
C	-5.79	-2.9	0.193	-15	0.000644	***
A	-4.56	-2.28	0.193	-11.8	0.00131	**
BD	3.7	1.85	0.193	9.57	0.00242	**
AC	3.6	1.8	0.193	9.32	0.00261	**
B	3.54	1.77	0.193	9.17	0.00274	**
AB	3.28	1.64	0.193	8.49	0.00343	**
D	-2.45	-1.23	0.193	-6.35	0.0079	**
CD	2.18	1.09	0.193	5.64	0.011	*
DE	1.93	0.963	0.193	4.98	0.0155	*
BE	1.49	0.747	0.193	3.87	0.0306	*
AD	-0.195	-0.0974	0.193	-0.504	0.649	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
BC	-0.131	-0.0657	0.193	-0.34	0.756	
CE	0.119	0.0596	0.193	0.309	0.778	

High mannose

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.51	0.756	0.155	4.89	0.0164	*
B	-1.39	-0.696	0.155	-4.5	0.0205	*
C	-0.446	-0.223	0.155	-1.44	0.245	
BC	0.369	0.185	0.155	1.19	0.318	
E	0.158	0.0791	0.155	0.511	0.645	
D	-0.139	-0.0693	0.155	-0.448	0.685	
AC	0.0897	0.0449	0.155	0.29	0.791	
AE	0.0894	0.0447	0.155	0.289	0.792	
DE	-0.0718	-0.0359	0.155	-0.232	0.831	
CE	-0.0552	-0.0276	0.155	-0.179	0.87	
BE	0.046	0.023	0.155	0.149	0.891	
AB	0.0432	0.0216	0.155	0.14	0.898	
AD	-0.0239	-0.012	0.155	-0.0774	0.943	
CD	-0.0232	-0.0116	0.155	-0.075	0.945	
BD	-0.000576	-0.000288	0.155	-0.00186	0.999	

Acidic variants

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-8.15	-4.08	0.239	-17	0.000441	***
A	-4.17	-2.08	0.239	-8.7	0.00319	**
B	2.97	1.48	0.239	6.19	0.00848	**
D	-2.55	-1.28	0.239	-5.34	0.0129	*
AB	1.98	0.992	0.239	4.15	0.0255	*
AC	-1.93	-0.964	0.239	-4.03	0.0275	*
BE	1.73	0.865	0.239	3.61	0.0364	*
AD	-0.669	-0.335	0.239	-1.4	0.256	
CE	-0.494	-0.247	0.239	-1.03	0.378	
CD	-0.492	-0.246	0.239	-1.03	0.379	
BD	-0.215	-0.108	0.239	-0.449	0.684	
BC	0.151	0.0757	0.239	0.316	0.773	
DE	0.0324	0.0162	0.239	0.0676	0.95	
E	0.0235	0.0117	0.239	0.049	0.964	
AE	0.0226	0.0113	0.239	0.0472	0.965	

Aggregates (HMW)

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.834	0.417	0.0207	20.1	0.000268	***
E	0.578	0.289	0.0207	13.9	0.0008	***
BE	-0.141	-0.0705	0.0207	-3.4	0.0424	*
C	0.116	0.058	0.0207	2.8	0.0678	
CD	-0.0742	-0.0371	0.0207	-1.79	0.171	
AD	-0.0666	-0.0333	0.0207	-1.61	0.206	
AB	-0.0593	-0.0297	0.0207	-1.43	0.248	
DE	-0.0554	-0.0277	0.0207	-1.34	0.274	
D	-0.0548	-0.0274	0.0207	-1.32	0.278	
BD	-0.0547	-0.0274	0.0207	-1.32	0.278	
AE	0.0416	0.0208	0.0207	1	0.39	
BC	-0.0235	-0.0118	0.0207	-0.567	0.61	
CE	0.0222	0.0111	0.0207	0.537	0.629	
B	-0.00671	-0.00336	0.0207	-0.162	0.882	
AC	-0.00663	-0.00332	0.0207	-0.16	0.883	

11.2 Response-surface coefficients — remaining CQAs

High mannose

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.89	0.163	36	2.06e-14	***
A	1.09	0.111	9.77	2.36e-07	***
B	-0.421	0.111	-3.78	0.00227	**
E	-0.188	0.111	-1.69	0.115	
C	0.212	0.111	1.91	0.0788	
AB	0.148	0.118	1.26	0.231	
AE	-0.29	0.118	-2.46	0.0289	*
AC	0.293	0.118	2.48	0.0276	*
BE	0.0874	0.118	0.74	0.472	
BC	-0.118	0.118	-0.996	0.337	
EC	-0.0402	0.118	-0.341	0.739	
A ²	0.116	0.294	0.393	0.701	
B ²	0.094	0.294	0.32	0.754	
E ²	-0.215	0.294	-0.731	0.478	
C ²	0.126	0.294	0.43	0.674	

Acidic variants

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	22	0.542	40.6	4.43e-15	***
A	-2.01	0.369	-5.44	0.000113	***

Term	Coef.	Std. err.	t	p-value	Sig.
B	1.9	0.369	5.14	0.000191	***
E	-0.124	0.369	-0.335	0.743	
C	-4.09	0.369	-11.1	5.4e-08	***
AB	1.11	0.392	2.84	0.0139	*
AE	-0.793	0.392	-2.03	0.0638	
AC	-0.248	0.392	-0.633	0.538	
BE	0.655	0.392	1.67	0.118	
BC	-0.0111	0.392	-0.0283	0.978	
EC	0.465	0.392	1.19	0.256	
A ²	-0.586	0.975	-0.601	0.558	
B ²	1.03	0.975	1.06	0.311	
E ²	-0.718	0.975	-0.736	0.475	
C ²	0.13	0.975	0.133	0.896	

Aggregates (HMW)

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	1.52	0.0341	44.8	1.25e-15	***
A	0.367	0.0232	15.8	7.26e-10	***
B	-0.0157	0.0232	-0.675	0.512	
E	0.281	0.0232	12.1	1.91e-08	***
C	0.0225	0.0232	0.971	0.349	
AB	0.0135	0.0246	0.547	0.593	
AE	-0.0106	0.0246	-0.43	0.674	
AC	0.0179	0.0246	0.727	0.48	
BE	-0.00713	0.0246	-0.29	0.777	
BC	0.00414	0.0246	0.168	0.869	
EC	0.0105	0.0246	0.426	0.677	
A ²	-0.0533	0.0613	-0.87	0.4	
B ²	-0.119	0.0613	-1.93	0.0752	
E ²	0.143	0.0613	2.33	0.0366	*
C ²	-0.0626	0.0613	-1.02	0.326	

Appendix D — Analytical methods summary

Table 11.13: Analytical methods, techniques and validation references (performance characteristics per the cited method validation reports).

Method	Technique	Analytes / CQAs	Validation ref.	Reportable
N-glycan map	2-AB HILIC-UPLC	Afucosylation, high mannose, galactosylation	AMV-3010	% of glycans
Size variants	SEC-HPLC	Aggregate (HMW), monomer	AMV-3011	% HMW
Host-cell protein	ELISA	HCP	AMV-3012	ng/mg
Charge variants	icIEF	Acidic / main / basic variants	AMV-3013	% acidic
Residual DNA	qPCR	Host-cell DNA	AMV-3014	ng/dose